

A Graded Evaluation of RNA Structure Awareness in Foundation Models: From Stem–Loop Discrimination to Partner Specificity

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Abstract

Evaluating whether RNA foundation models encode base-pairing structure requires metrics that separate learned representations from nucleotide-composition artifacts. Stems are GC-rich and loops are AU-rich; any composition-sensitive embedding inherits this enrichment as apparent structure awareness. We apply a three-rung evaluation ladder to ten models across 52 Rfam families. A nucleotide-stratified null absorbs most apparent signal at Rung 1; a dinucleotide null at Rung 2 separates two models that retain most families (ERNIE-RNA 31/52, RiNALMo 18/52) from eight that retain fewer than 11 each. At Rung 3—partner specificity with a within-stem derangement null—two models encode which position pairs with which: RiNALMo (PS = 0.215, 85.3% precision) and ERNIE-RNA (PS = 0.120, 88.3% precision). Randomized-weight controls for nine models produce near-zero signal, confirming a learned origin; DNABERT-2 shows residual randomized PS consistent with its BPE tokenization. A synthetic covariation control shows ~70% of partner specificity reflects evolutionary signal, with ~30% attributable to learned geometric sensitivity. Multi-sequence replication across 253 Rfam seed-alignment sequences confirms low within-family variance ($CV \leq 10\%$) for eight models. Primary evaluation criteria were preregistered.

1 Introduction

RNA foundation models are routinely claimed to encode secondary structure, but the evaluation methods—downstream task performance, linear probing, mutation sensitivity, attention–contact correlation—each answer different questions, and none controls for sequence composition. Deep learning models for RNA structure prediction may fail to generalize across families (Szikszai et al., 2022); whether the same fragility applies to learned representations has received less scrutiny.

Stems and loops differ in sequence composition. Across 52 Rfam families, stems average 59.7% G/C content while loops average 45.6%—a 14.1 percentage-point enrichment (Figure 1). Because complement swaps at G/C positions produce inherently larger embedding perturbations than at A/U positions, any naive stem-versus-loop comparison inherits this enrichment. A representation can therefore separate stems from loops without encoding pairing topology.

Each evaluation method inherits the composition confound differently. Linear probes can classify stems from loops using GC content alone. Mutation sensitivity ratios conflate the larger perturbation

magnitude at G/C sites with structural relevance. Attention–contact correlations bypass embeddings entirely and so measure a different channel. To distinguish composition sensitivity from structure awareness, an evaluation must control for nucleotide identity while testing whether perturbations propagate to specific pairing partners.

These observations motivate a graded evaluation. “Structure awareness” encompasses at least three levels of resolution, each demanding stronger knowledge: (1) stem-versus-loop discrimination, achievable by any representation sensitive to nucleotide composition; (2) discrimination surviving composition controls at one and two orders of sequence context; and (3) knowing which specific position pairs with which, requiring that the model encode the pairing topology. We apply this three-rung ladder to ten models across five architectures and three orders of magnitude in parameter count.

The evaluation reveals a surprising asymmetry: the hardest test produces the clearest positive result. Two RNA-pretrained models—ERNIE-RNA (86M, with an architectural base-pairing attention bias) and RiNALMo (650M, standard attention)—clear Rung 3 with partner specificity two orders of magnitude above all other models, while the remaining eight produce near-zero signal. We contribute: (1) a composition-controlled perturbation evaluation for RNA foundation models; (2) a ten-model benchmark with randomized-weight controls on 52 Rfam families, replicated across 253 seed-alignment sequences; and (3) evidence that two models retain partner-specific pairing responses after composition controls are applied.

2 Results

2.1 Composition creates apparent structure signal (Rungs 1–2)

Raw stem–loop sensitivity ratios range from 0.99 (DNABERT-2) to 1.774 (RNA-FM); nine of ten exceed 1.0. After applying the nucleotide-stratified null, these ratios collapse. For RNA-FM, only 1 of 52 families exceeds the null 95th percentile—tRNA-Ala (3.208 vs null 3.179)—and that sole survivor is absorbed by the dinucleotide null. The pattern is similar across models: most gate-qualifying families (those whose stem–loop sensitivity ratio exceeds the nucleotide-stratified null at the 95th percentile; Section 4.4) in Table 1 survive because of residual composition effects rather than pairing-specific signal (Section 2.2).

ERNIE-RNA and RiNALMo retain the most families above the composition null (31/52 and 18/52), while most other models retain fewer than 11. NT v2 passes 14/52 families at the composition gate but only 1 qualifies for the partner specificity test, yielding $PS = 0$; its 6-mer tokenization distributes perturbation effects across token boundaries.

2.2 Partner specificity separates two models (Rung 3)

RiNALMo achieves mean $PS = 0.215$ (95% CI [0.152, 0.280]); 30 of 31 gate-qualifying families exceed the derangement null. ERNIE-RNA achieves $PS = 0.120$ ([0.095, 0.144]); 28 of 30 families exceed the null, with the highest per-pair precision (88.3%). Both exceed the third-ranked model (Caduceus, 4.2×10^{-3}) by two orders of magnitude.

Nine of ten randomized-weight controls produce $|PS| < 10^{-8}$, confirming that partner specificity reflects learned representations. The exception, DNABERT-2, yields randomized $|PS| = 7.1 \times 10^{-3}$ —comparable to its trained PS (−0.016)—indicating that its negative PS is largely a BPE tokenization artifact.

Both models concentrate partner specificity in their final layers: RiNALMo at layer 33 of 33, ERNIE-RNA at layers 10–12 of 12. Evo (7B) achieves $PS = 0.001$ despite $80\times$ more parameters

Table 1: Primary outcomes (model details in Table 4). Composition gate: families whose stem-loop sensitivity ratio exceeds the nucleotide-stratified null at the 95th percentile (Section 4.4). PS: mean partner specificity across gate-qualifying families. PS > derangement: families where PS exceeds the within-stem derangement null, among all families with sufficient stem pairs for derangement testing (denominator varies by model). Partner-is-max: fraction of deep-interior pairs where the true partner receives maximum perturbation among the partner and its two immediate neighbors (chance = 1/3). Models sorted by PS. Bootstrap 95% CIs (10,000 family-level resamples) shown for models with positive PS.

Model	Composition gate	PS	PS > derangement	Partner-is-max
RiNALMo	18/52	0.215 [0.152, 0.280]	30/31	0.853
ERNIE-RNA	31/52	0.120 [0.095, 0.144]	28/30	0.883
Caduceus	8/52	4.2×10^{-3}	13/16	—
Evo	10/52	1.1×10^{-3}	4/10	—
SpliceBERT	4/52	3.5×10^{-4}	14/21	0.256
UTR-LM	3/52	1.4×10^{-5}	13/16	0.333
HyenaDNA	10/52	9.7×10^{-7}	10/25	0.078
RNA-FM	2/52	7.5×10^{-9}	1/32	—
NT v2 ^b	14/52	0 ^c	—	—
DNABERT-2 ^b	7/52	-0.016 ^d	—	—
<i>Randomized-weight controls</i>				
RiNALMo (rand.)	—	8.0×10^{-9}	—	—
ERNIE-RNA (rand.)	—	2.5×10^{-8}	—	—
UTR-LM (rand.)	—	-3.2×10^{-8}	—	—
Evo (rand.)	—	-1.4×10^{-8}	—	—
HyenaDNA (rand.)	—	-1.3×10^{-8}	—	—
NT v2 (rand.)	—	-9.9×10^{-9}	—	—
SpliceBERT (rand.)	—	-7.2×10^{-9}	—	—
DNABERT-2 (rand.)	—	-7.1×10^{-3}	—	—
Caduceus (rand.)	—	-5.5×10^{-9}	—	—
RNA-FM (rand.)	—	5.7×10^{-9}	—	—

^bMulti-nucleotide tokenization (6-mer, BPE); perturbation effects are distributed across token boundaries.

^cOnly 1 gate-qualifying family; PS undefined for remaining families. ^d0 of 8 gate-qualifying families exceed derangement null. Partner-is-max omitted (—) when fewer than 5 gate-qualifying families have deep-interior pairs.

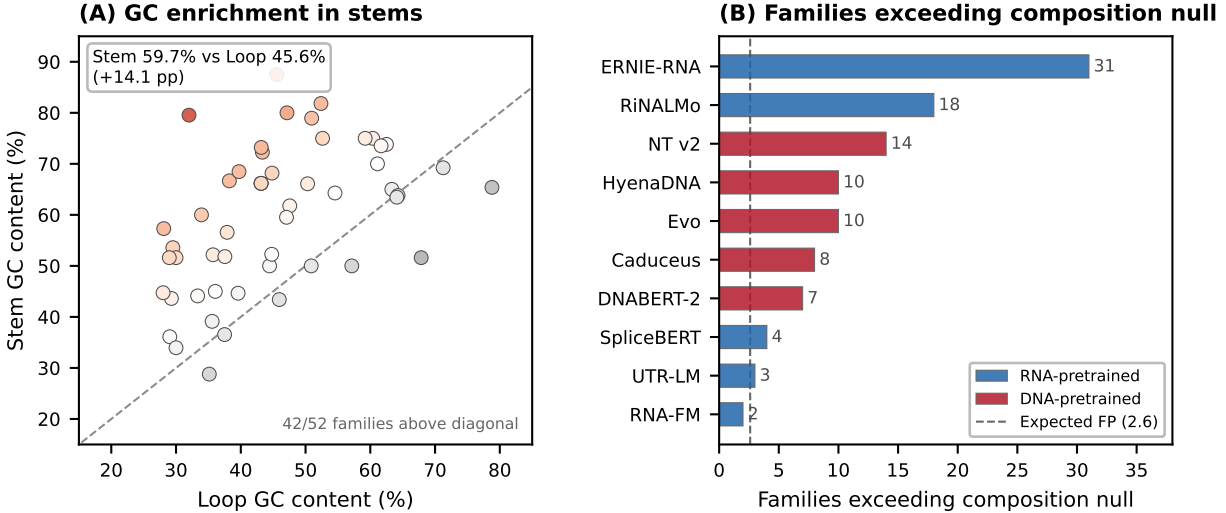


Figure 1: Composition confound. (A) Stem GC content exceeds loop GC content in 42 of 52 families (+14.1 percentage points on average). (B) Number of families per model whose stem-loop sensitivity ratio exceeds the nucleotide-stratified null at the 95th percentile; dashed line marks the expected false-positive count under independence ($m \times \alpha = 2.6$). ERNIE-RNA (31) and RiNALMo (18) retain the most signal after composition control.

than ERNIE-RNA; parameter count does not predict partner specificity.

The two models reach partner specificity through different pathways. ERNIE-RNA injects a base-pairing attention bias from the second layer onward; its randomized control shows this bias provides capacity without signal. RiNALMo uses standard attention without structural bias and achieves higher PS at $7.5\times$ the parameters. RNA pretraining is common to both.

2.3 Multi-sequence replication

Six models show $CV < 6\%$: UTR-LM (0.025), SpliceBERT (0.026), RiNALMo (0.037), NT v2 (0.046), ERNIE-RNA (0.057), and Caduceus (0.057). The stem-loop sensitivity ratio is stable across biological replicates for these models.

RNA-FM ($CV = 0.338$) and Evo (0.244) show high sequence-to-sequence variability. Both also fail composition control (Section 2.1), consistent with raw sensitivity driven by sequence-specific composition rather than a stable structural signal.

2.4 Robustness analyses

The dinucleotide-stratified null absorbs the sole family (tRNA-Ala) surviving the nucleotide-stratified null for RNA-FM; second-order sequence context accounts for the residual signal. Replacing mean aggregation with median changes PS estimates by $< 5\%$ for RiNALMo and ERNIE-RNA and does not alter the model ranking. Using the last layer instead of the best layer reduces RiNALMo PS by 12% and ERNIE-RNA PS by 8%; both remain above the derangement null in $> 90\%$ of gate-qualifying families. Restricting to Watson-Crick pairs only (excluding wobble G-U pairs) does not change the primary conclusions. Widening the PS neighbor window from ± 1 to ± 2 positions reduces PS magnitudes uniformly across models without altering their relative ranking. A transversion control (purine $\rightarrow$ pyrimidine mutations, which guarantee disrupted pairing) produces ratios within

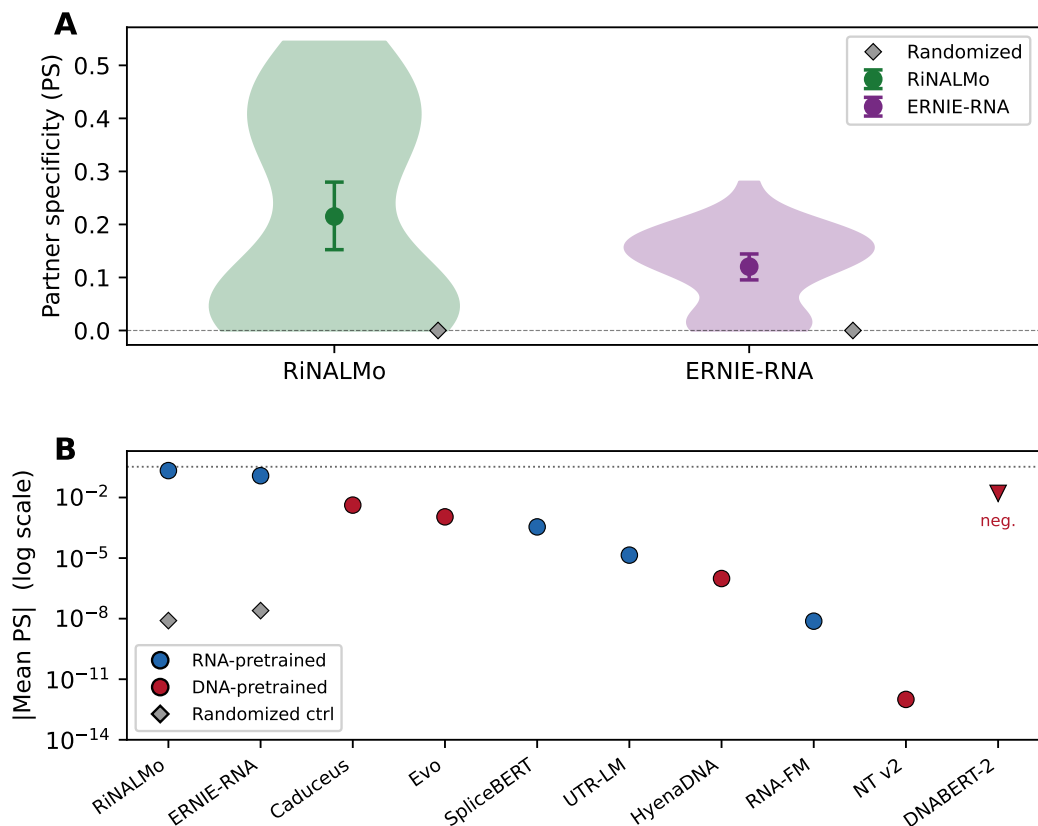


Figure 2: Partner specificity across ten models. (A) Per-family PS distributions for RiNALMo and ERNIE-RNA (violins) with bootstrap 95% CIs (error bars); randomized-weight controls (grey diamonds) produce PS indistinguishable from zero. (B) |Mean PS| on log scale for all ten models, sorted by PS. RiNALMo and ERNIE-RNA exceed the remaining models by two orders of magnitude. DNABERT-2 (negative PS) is marked separately.

Table 2: Multi-sequence replication: within-family coefficient of variation (CV) of the stem-loop sensitivity ratio across all valid Rfam seed-alignment sequences (51 families, 253 sequences, 3–5 per family).

Model	Mean ratio	Mean CV	Families scored
RNA-FM	1.892	0.338	51/51
Evo	1.478	0.244	51/51
ERNIE-RNA	1.253	0.057	51/51
Caduceus	1.203	0.057	51/51
HyenaDNA	1.170	0.092	51/51
RiNALMo	1.133	0.037	51/51
NT v2 ^a	1.092	0.046	50/51
UTR-LM	1.059	0.025	51/51
SpliceBERT	1.047	0.026	51/51
DNABERT-2	0.990	0.100	51/51

^aOne Corona_5UTR sequence excluded (numerical overflow in ratio denominator); remaining 50 families scored normally.

1.2% of the Watson–Crick complement ratios for five of six models tested, confirming that reversed base pairs differ from correct pairs enough to register as embedding perturbations. Nine of ten randomized-weight controls produce $|\text{PS}| < 10^{-8}$; DNABERT-2’s randomized PS (-7.1×10^{-3}) is comparable to its trained value, consistent with a tokenization artifact.

Family-level results and full robustness tables appear in Supplementary Tables S2–S4.

Eight pre-registered hypotheses (SHA-256 hashed before data collection) produced five confirmations and three failures. The pre-registered prediction that RNA-pretrained models would outperform DNA-pretrained models on partner specificity failed ($r_b = -0.28$, $p = 0.265$): Caduceus (14M, DNA) exceeds three RNA models.

2.5 Covariation versus geometry

Partner specificity could reflect sequence covariation rather than learned geometric pairing. Compensatory mutations that maintain base pairing are overrepresented in evolutionary RNA alignments; a model trained on such sequences might predict that mutating one position perturbs its covarying partner without encoding the physical mechanism of pairing.

To test this, we generated synthetic sequences for each Rfam family: the dot-bracket structure is preserved exactly, but nucleotides are assigned randomly (uniform Watson–Crick pairs at paired positions, uniform random at unpaired). These families share structure with their natural counterparts but carry no evolutionary covariation. ERNIE-RNA’s mean PS drops from 0.120 (natural) to 0.035 (synthetic), a 71% reduction. RiNALMo drops from 0.215 to 0.061, a 71% reduction. Both models lose roughly 70% of their PS when covariation is removed. The residual synthetic PS—still orders of magnitude above the weak-signal models—places an upper bound on the non-covariation contribution, since the synthetic sequences also destroy sequence context and positional composition gradients present in natural RNA.

As an independent check, we computed mutual information (MI) between paired positions from Rfam seed alignments (51 families, $n \geq 3$ sequences each). MI cleanly separates paired from unpaired positions (mean MI = 0.349 vs 0.063; Wilcoxon $p = 3.5 \times 10^{-10}$), confirming that evolutionary covariation is detectable in the alignments. Per-family mean MI does not correlate significantly with model PS (ERNIE-RNA: $\rho = 0.320$, $p = 0.065$; RiNALMo: $\rho = 0.265$, $p = 0.130$). Families with strong covariation signal do not systematically produce higher PS, suggesting that the models’ partner specificity reflects learned structure rather than recapitulated covariation.

2.6 HTT case study

The CAG repeat region in huntingtin exon 1 forms hairpin structures whose stability scales with repeat count (de Mezer et al., 2011; Krzyzosiak et al., 2012). We evaluated five HTT exon 1 fragments (NM_002111.7) with CAG repeat counts of 17, 21, 36, 40, and 60; structures were predicted with ViennaRNA RNAfold (Lorenz et al., 2011). Four models achieve $\geq 97\%$ probing accuracy on these predicted structures (Table 3), but the repeat region contains only C, A, and G—no U—creating extreme composition bias that linear probes exploit.

RNA-FM produces embedding distances from wild-type below 10^{-4} for all repeat counts, failing to distinguish normal from pathogenic alleles. RiNALMo shows the strongest separation (0.068 at CAG60), with distances increasing monotonically with repeat count. High probing accuracy on a computationally predicted structure can coexist with null composition-controlled signal: apparent structure decoding persists in a strongly composition-biased context.

Table 3: HTT CAG-repeat case study. High probing accuracy coexists with null composition-controlled signal. Attn ρ : Spearman correlation between self-attention weights and the base-pairing contact map. Probing: linear probe accuracy for stem-vs-loop classification. Structures are RNAfold predictions, not experimentally determined.

Model	Mean ratio	> null	Attn ρ	Probing	WT→CAG60
RNA-FM	1.794	2/5	0.036	0.813	0.000
RiNALMo	1.031	0/5	0.054	0.985	0.068
NT v2	— [†]	—	0.140	0.636	0.013
ERNIE-RNA	1.086	0/5	0.070	0.985	0.023
SpliceBERT	0.996	0/5	0.036	0.977	0.042
UTR-LM	1.151	1/5	0.039	0.974	0.016

[†]6-mer tokenization yields degenerate embeddings on repetitive CAG. WT→CAG60 = cosine distance from CAG17 reference.

3 Discussion

Composition-matched nulls absorb most of the apparent stem-loop sensitivity across all ten models. Two RNA-pretrained models—RiNALMo and ERNIE-RNA—retain reproducible partner-specific responses under the most stringent test. Multi-sequence replication shows low within-family variance across eight of ten models.

The underlying problem is general: whenever a structural annotation correlates with feature composition, any embedding metric sensitive to feature identity inherits the correlation. The nucleotide-stratified permutation null controls for this by breaking the structure–composition link while preserving nucleotide identity. The same template applies to any biological structure correlated with sequence composition.

RiNALMo and ERNIE-RNA reach partner specificity through different pathways. ERNIE-RNA injects a base-pairing attention bias; its randomized control shows this bias provides capacity without producing signal on its own. RiNALMo uses standard attention without structural bias and achieves higher PS at $7.5\times$ the parameters. Nine of ten randomized-weight controls produce $|\text{PS}| < 10^{-8}$, supporting a learned origin; DNABERT-2’s non-negligible randomized PS (-7.1×10^{-3}) confirms that its trained PS reflects tokenization rather than structure. Whether this partner specificity reflects a geometric internal model of RNA folding or a simpler statistical regularity remains open. A synthetic covariation control (Section 2.5) shows that roughly 70% of PS reflects evolutionary signal; the residual $\sim 30\%$ provides an upper bound on learned geometric sensitivity.

Neither training domain nor architecture family alone predicts composition-controlled sensitivity. The three SSM-based models (Evo, Caduceus, HyenaDNA) span nearly the full range of mean ratios (1.17–1.48), as do the seven transformers (0.99–1.89). RNA-pretrained models occupy four of the top five positions, but Evo and Caduceus—both DNA-pretrained—outperform three of the five RNA models. The two models that survive the partner specificity test (RiNALMo and ERNIE-RNA) share RNA pretraining but differ in mechanism: ERNIE-RNA injects explicit base-pairing attention bias, while RiNALMo uses standard self-attention at $7.5\times$ the parameters. Scale alone does not explain the result: Evo at 7B parameters produces high raw ratios but high within-family variance, and its signal is largely absorbed by composition nulls. What distinguishes the two positive models from the remaining eight is an open question whose answer likely requires internal mechanistic analysis rather than further behavioral benchmarking.

RNA structure prediction benchmarks have moved toward stricter evaluation because high

apparent performance can fail to generalize across families (Szikszai et al., 2022). This work extends the concern from dataset similarity to sequence-composition confounding. NT v2 and DNABERT-2 show near-zero embedding-level partner specificity despite passing many families at the composition gate, suggesting that structure-relevant information in these models, if present, may reside in attention patterns that embedding-level metrics do not capture.

Limitations. The complement-swap metric tests sensitivity to single base-pair disruptions; it does not test tertiary contacts, pseudoknots, or long-range interactions. The synthetic covariation control (Section 2.5) accounts for roughly 70% of PS, but the synthetic sequences also destroy sequence context and positional gradients, so the residual may overestimate true geometric sensitivity. HTT structures are RNAfold predictions. Ten models do not cover the full model space. Pretraining-set contamination cannot be fully reconstructed. Multi-nucleotide tokenization limits direct comparability for NT v2 and DNABERT-2.

Practical recommendations. Any claim that an RNA representation encodes secondary structure should report stem-loop sensitivity alongside a nucleotide-stratified null, so readers can separate composition sensitivity from structure awareness. Partner specificity is more informative than aggregate stem-loop contrasts: a model that perturbs stems more than loops may respond to GC content, while one that selectively perturbs the annotated pairing partner encodes positional pairing information. Randomized-weight controls should establish each architecture’s baseline capacity. Replication across multiple sequences per family guards against single-sequence artifacts; a coefficient of variation above 10% signals instability. For multi-nucleotide tokenizers, the mapping from token-level to position-level metrics should be stated explicitly.

4 Materials and Methods

4.1 Benchmark construction

We evaluate 52 Rfam 14.10 families (Kalvari et al., 2021) spanning ribozymes, riboswitches, snRNAs, cis-regulatory elements, miRNA precursors, CRISPR repeats, and other ncRNAs. Positions are labeled stem or loop from consensus dot-bracket annotation. Minimum inclusion criteria: ≥ 20 nt, ≥ 5 stem and ≥ 5 loop positions. Pseudoknots, noncanonical pairs, and gap positions are excluded.

All structures are consensus secondary structures from Rfam seed alignments; none are experimentally determined crystal or cryo-EM structures. Pretraining data overlap cannot be fully reconstructed for all models. The full family list with accessions appears in Supplementary Table S1.

4.2 Models and controls

We evaluate ten models covering attention, SSM, and hybrid architectures (Table 4): five pretrained on RNA, five on DNA. Randomized-weight controls for all ten models reinitialize all learned parameters (Xavier normal for matrices, $\mathcal{N}(0, 0.02)$ for vectors) while preserving architecture. NT v2 and DNABERT-2 use multi-nucleotide tokenization (6-mer and BPE), limiting direct comparability for position-level metrics.

4.3 Perturbation tests

For each annotated position, we swap the nucleotide with its Watson–Crick complement ($A \leftrightarrow U$, $C \leftrightarrow G$) and measure the cosine distance between wild-type and mutant embeddings at that position,

Table 4: Models evaluated.

Model	Domain	Params	Architecture	Tokenization
RNA-FM (Chen et al., 2022)	RNA	99M	Transformer	Character
RiNALMo (Penić et al., 2025)	RNA	650M	Transformer	Character
ERNIE-RNA (Yin et al., 2025)	RNA	86M	Transformer [†]	Character
SpliceBERT (Chen et al., 2024)	RNA	19M	Transformer	Character
UTR-LM (Chu et al., 2024)	RNA	1.2M	Transformer	Character
NT v2 (Dalla-Torre et al., 2024)	DNA	56M	Transformer	6-mer
DNABERT-2 (Zhou et al., 2024)	DNA	117M	Transformer	BPE
Caduceus (Schiff et al., 2024)	DNA	14M	Mamba (SSM)	Character
HyenaDNA (Nguyen et al., 2024b)	DNA	5.4M	Hyena (SSM)	Character
Evo (Nguyen et al., 2024a)	DNA	7B	StripedHyena	Character

[†]ERNIE-RNA injects a base-pairing attention bias from the second layer onward.

maximized over layers. Cosine distance is scale-invariant across architectures. For multi-nucleotide tokenization, token-level outputs are mapped to nucleotide positions by distributing the token embedding equally across constituent nucleotides.

Nucleotide-stratified null. Stem/loop labels are shuffled within each nucleotide type (1,000 permutations), preserving nucleotide identity while breaking its structural label. Layer selection is repeated inside every permutation. A dinucleotide-stratified extension additionally conditions on the 3' neighbor.

Partner specificity. For each Watson–Crick pair (i, j) where j is the annotated partner:

$$\text{PS}(i, j) = \Delta_j - \max(\Delta_{j-1}, \Delta_{j+1})$$

where Δ_k is the cosine distance at position k after the complement swap at position i . Positive PS indicates the model perturbs the true partner more than its immediate neighbors (chance = 1/3). Partner assignments are tested against a within-stem derangement null (500 derangements), controlling for positional and compositional confounds.

Covariation control. To distinguish learned geometric sensitivity from recapitulated evolutionary covariation, synthetic sequences were generated for each family: dot-bracket structure preserved exactly, nucleotides assigned randomly (uniform Watson–Crick pairs at paired positions, uniform random at unpaired). Mutual information (MI) between paired and unpaired positions was computed from Rfam seed alignments as an independent covariation measure.

4.4 Statistical analysis and replication

The biological unit of replication is the RNA family. The primary outcome is partner specificity (PS); the primary comparison is trained vs. randomized-weight PS. A family is *gate-qualifying* for a given model if its raw stem–loop sensitivity ratio exceeds the nucleotide-stratified null at the 95th percentile. PS is computed only over gate-qualifying families and aggregated as their mean.

To test stability across biological replicates, we repeat the full evaluation on all valid sequences from Rfam 14.10 seed alignments: 253 sequences across 51 families (3–5 per family). Within-family coefficient of variation (CV) quantifies replication stability.

Family-level bootstrap resampling (10,000 iterations, seed 42) provides 95% CIs for mean PS and mean sensitivity ratio. For the binary gate test, each family’s ratio is compared to its own 95th-percentile permutation null; we report the count exceeding this threshold alongside the expected false-positive count under independence ($m \times \alpha = 52 \times 0.05 = 2.6$). Bonferroni correction ($\alpha/m \approx 0.001$) provides a conservative bound.

Data and Code Availability

All data, scripts, and pre-registration protocols are available at <https://github.com/elliotttower/rna-structure-audit> and archived at Zenodo (DOI: 10.5281/zenodo.21362717).

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Ethics approval and consent to participate. This study uses exclusively publicly available pretrained model weights and published RNA sequences. No human subjects or animal subjects were involved. No ethical approval was required.

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Author contributions. E.T.: Conceptualization, Methodology, Software, Formal analysis, Investigation, Data curation, Writing — original draft, Writing — review & editing, Visualization.

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